

COMP2511

Tute02



Agenda

- Code Review
- JavaDoc & Commenting
- Inheritance & Polymorphism
- toString & Equals
- Access Modifiers & Packages
- Assignment i!!!



Code Review



Code Review

super vs this

- What is the difference between super and this?
- What about super(...) and this(...)?
- What will the main code in `Rectangle` print and why?



Code Review

Abstract Classes vs Interfaces

- What is the purpose of using an abstract class in this code?
- What are some downsides of the use of the abstract class here?
- What is the difference between an abstract class and an interface? Why would you use one or the other?



Code Review

static

- Can you override a static method?
- What is the output of running `r2.getArea()` in main?
- What is the output of running `Shape.getCount()` in main?



JavaDoc & Commenting



- What are the main features of the JavaDoc present in Employee.java?
 - What is meant by the term "self-documenting code"?
- 



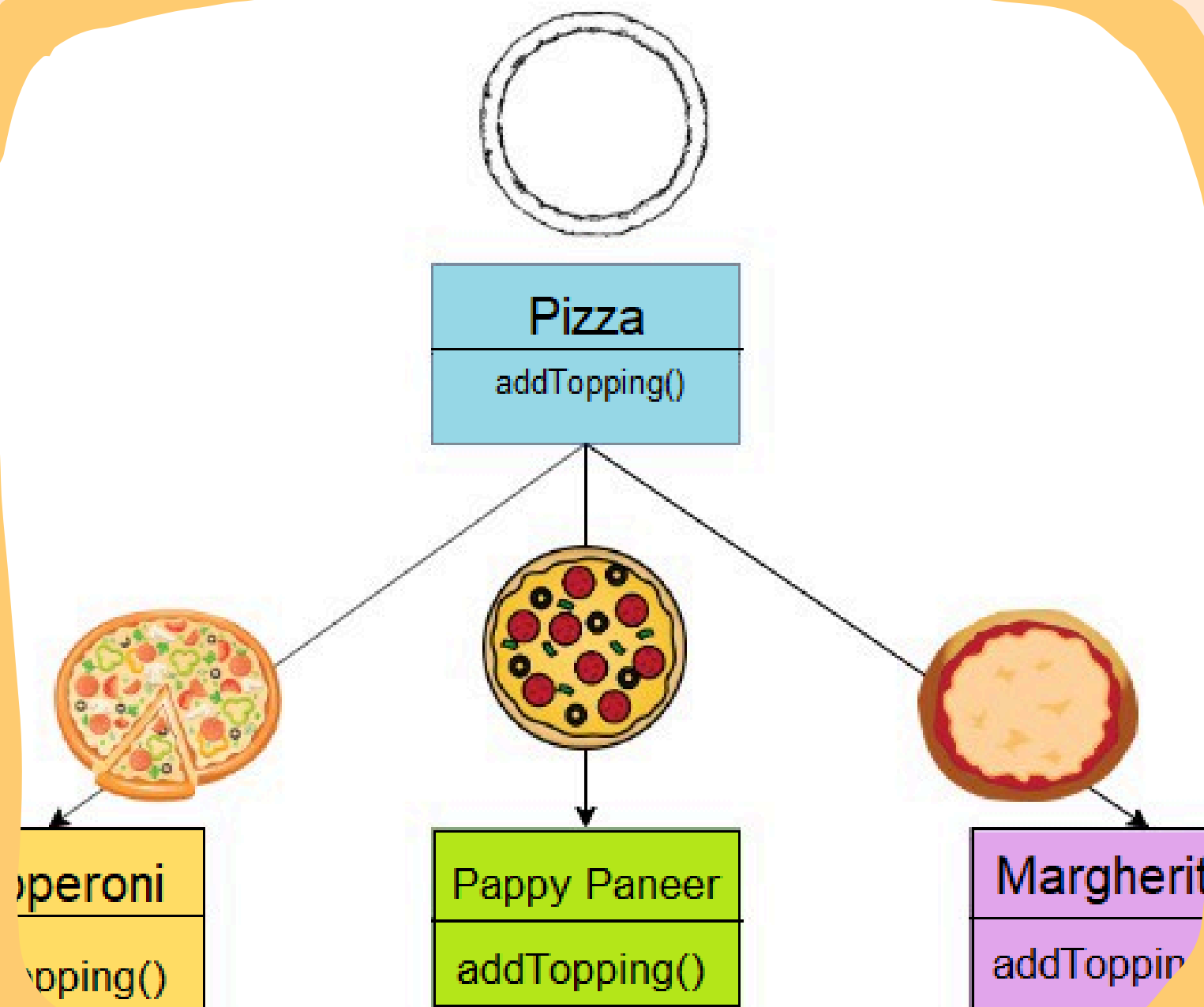
Why Use JavaDoc & Comments?

And when not to?

Inheritance & Polymorphism

Create a Manager subclass of Employee, that includes hire date.

- What constructors are appropriate?
- Is it appropriate to have a getter for the hire date? What about a setter?
- Why might adding certain getters and setters be bad design?



toString

Override the toString() method inherited from Object in both classes.

- What should the result of toString() contain?
- Does the method in Manager call the one in Employee?



The == Operator

1

What is a primitive type?

2

How are objects stored in Java?

3

Pass by reference vs pass by value?

The == Operator

4

How does ==
work on objects
vs primitives?

5

What does it
mean for
objects to be
equal?

6

When are a
super type and
a sub type
equal?

equals

Override the equals() method inherited from Object in both classes.

- What key things should the equals() method do?
- How should the type of the input object be checked? How should it be compared to the type of the calling object?
- How can the method in Manager utilise code in Employee to avoid repetition?



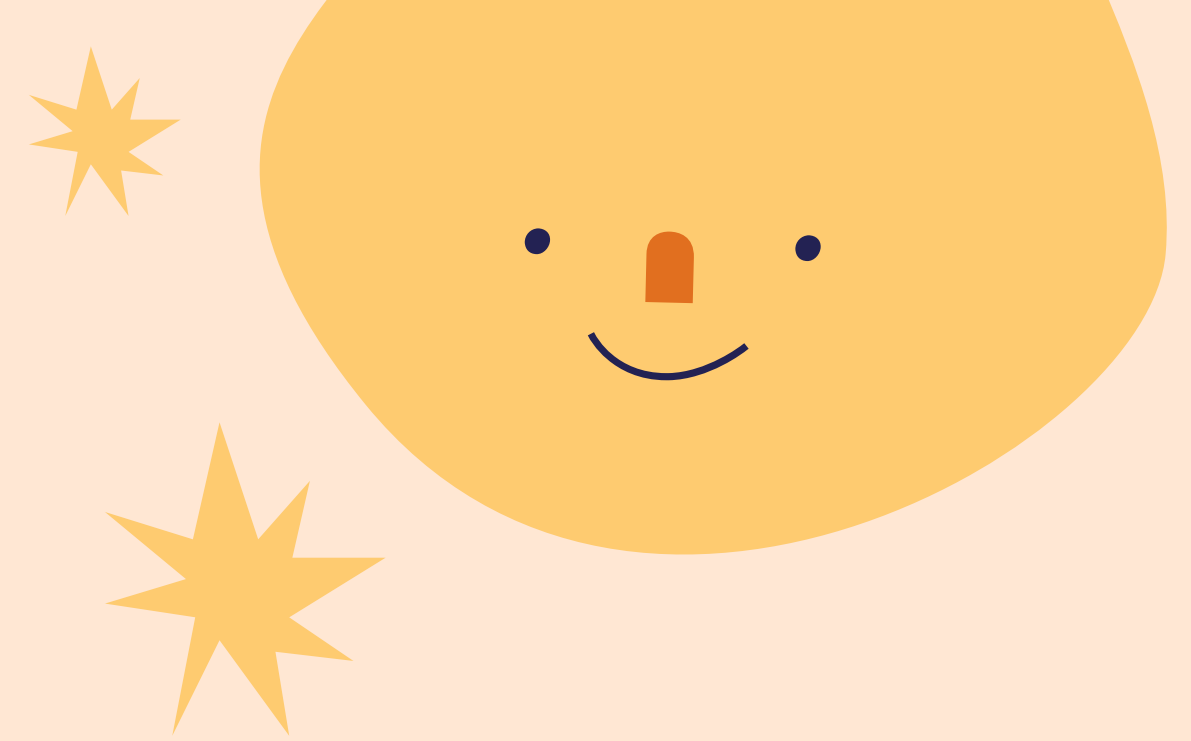
Access Modifiers & Packages



Access Modifiers

Modifier	Class	Package	Subclass	Global
Public	✓	✓	✓	✓
Protected	✓	✓	✓	✗
Default	✓	✓	✗	✗
Private	✓	✗	✗	✗

ASSIGNMENT i!!!



- Assignment i IS OUT THIS FRIDAY! - so take a look :D
- We will go through assignment in detail in Wk3 Lecture!
- Start it early!!! - maybe wait until after Wk3 Lecture if you don't understand the assignment well
- REMEMBER THIS IS A DESIGN COURSE!!! PLAN OUT YOUR DESIGN BEFORE YOU START CODING!!!